NEET UG 2027: Organic Chemistry Master Compendium

Reaction Mechanisms, Named Conversions, Reagents & Distinction Tests

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1 General Organic Chemistry (GOC) & Electronic Effects

Fundamental Electronic Effects in Organic Molecules

Organic reaction mechanisms and intermediate stabilities are governed by four fundamental displacement effects:

1. **Inductive Effect (I-Effect):** Permanent polarization through σ-bonds due to electronegativity difference. Decreases rapidly with distance ($I \propto 1/d^3$, negligible beyond 3 carbons).
   - $-I$ **Order:** $-NF_3^+ > -NR_3^+ > -NO_2 > -CN > -COOH > -F > -Cl > -Br > -I > -OR > -OH > -C \equiv CH > -NH_2 > -C_6H_5 > -CH = CH_2 > -H$
   - $+I$ **Order:** $-O^- > -COO^- > -CR_3(3°) > -CHR_2(2°) > -CH_2R(1°) > -CH_3 > -T > -D > -H$
2. **Electromeric Effect (E-Effect):** Temporary complete transfer of a shared pair of π-electrons to one of the atoms in the presence of an attacking reagent.
   - $+E$ **Effect:** π-electrons transferred to the atom where the electrophile attacks (e.g., addition of H^+ to alkenes).
   - $-E$ **Effect:** π-electrons transferred to the atom from which the nucleophile moves away (e.g., addition of CN^- to carbonyls).
3. **Resonance / Mesomeric Effect (R/M-Effect):** Permanent delocalization of conjugated π-electrons or lone pairs.
   - $+M$ **(Electron Donating via Lone Pair):** $-O^- > -NH_2 > -NHR > -OH > -OR > -NHCOR > -OCOR > -Ph > -X$ ($F > Cl > Br > I$).
   - $-M$ **(Electron Withdrawing via Multiple Bonds):** $-NO_2 > -CN > -SO_3H > -CHO >> C=O > -COOH > -COOR > -CONH_2$.
4. **Hyperconjugation (Baker-Nathan Effect):** Delocalization of σ_{C-H} electrons with an adjacent empty p-orbital or π^* orbital.

$$\text{Number of hyperconjugative structures} = \text{Number of } \alpha\text{-H atoms} \tag{1}$$

Stability Hierarchy of Reactive Intermediates

Carbocations ($sp^2, 6e^-$): Tropylium > CPM > Ph_3C^+ > Allyl/Benzyl $\approx 3° > 2° > 1° > CH_3^+$
Stabilized by: $+M >$ Hyperconjugation (α-H) $> +I$.
Destabilized by: $-M, -I$.

Carbanions (sp^3 or $sp, 8e^-$): $CH \equiv C^- > CH_2 = CH^- > CH_3^- > 1° > 2° > 3°$
Stabilized by: $-M, -I$, high % s-character, aromaticity.

Free Radicals ($sp^2, 7e^-$): $Ph_3C^\bullet >$ Allyl/Benzyl $> 3° > 2° > 1° > CH_3^\bullet$
Governed by hyperconjugation and resonance.

Carbocation Rearrangements: Hydride, Alkyl Shift & Ring Expansion

Carbocations spontaneously undergo 1,2-shifts to achieve higher thermodynamic stability:

- **Migratory Aptitude in Carbocations:** H^- > Aryl (Ar) > Alkyl ($3° > 2° > 1° > CH_3$).
- **Ring Expansion:** 4-membered ring carbocation → 5-membered ring (relieves angle strain from 90° → 108°); 5-membered ring carbocation → 6-membered ring (relieves strain to stable chair conformation).

- **Ring Contraction:** 7-membered ring carbocation undergoes contraction to stable 6-membered ring.

Quantitative Acidity & Basicity Sequences

1. Acidity Orders (Stabilization of Conjugate Base A^-):

General Class:	$R-SO_3H > R-COOH > H_2CO_3 > Ph-OH > H_2O > R-OH > HC\equiv CH > NH_3 > RH$
Haloacetic Acids:	$CF_3COOH > CCl_3COOH > CHCl_2COOH > CH_2FCOOH > CH_2ClCOOH$
Substituted Phenols:	Picric acid > *p*-nitrophenol > *o*-nitrophenol > *m*-nitrophenol > Phenol
Ortho-Effect:	Any ortho-substituted benzoic acid is stronger than benzoic acid.

2. Basicity Orders of Amines (Availability of Lone Pair on N):

Gas Phase:	3° Amine > 2° Amine > 1° Amine > NH_3 (strictly $+I$ governed)
Aqueous (Methyl):	$(CH_3)_2NH(2^\circ) > CH_3NH_2(1^\circ) > (CH_3)_3N(3^\circ) > NH_3$ **[Trick: 213]**
Aqueous (Ethyl):	$(C_2H_5)_2NH(2^\circ) > (C_2H_5)_3N(3^\circ) > C_2H_5NH_2(1^\circ) > NH_3$ **[Trick: 231]**
Aromatic Amines:	Aliphatic amines $\gg NH_3$ > *p*-toluidine > Aniline > *p*-nitroaniline.

NEET Trap: Ortho-Effect, H-Bonding & Volatility

- ***o*-Nitrophenol vs *p*-Nitrophenol:** *o*-nitrophenol possesses **intramolecular H-bonding** (chelation), making it steam volatile with lower boiling point and less water solubility. *p*-nitrophenol possesses **intermolecular H-bonding**, giving it a higher boiling point and higher acidity ($pK_a = 7.15$ vs 7.23 for ortho due to absence of steric repulsion).
- **Picric Acid ($pK_a = 0.38$):** Stronger than carbonic acid and even many carboxylic acids; readily gives effervescence with aqueous $NaHCO_3$.
- **Phenol with $NaHCO_3$:** Phenol does NOT evolve CO_2 with $NaHCO_3$ (pK_a phenol $\approx 10 > pK_a$ $H_2CO_3 \approx 6.4$). Nitrophenols with ≥ 2 nitro groups DO dissolve in $NaHCO_3$.

2 Hydrocarbons: Alkanes, Alkenes, Alkynes & Arenes

Alkane Synthesis & Free Radical Transformations

1. Wurtz Reaction:

$$2R-X + 2Na \xrightarrow{\text{dry ether}} R-R + 2NaX \quad \text{(Free radical mechanism; best for symmetrical alkanes with even C)} \tag{2}$$

2. Decarboxylation (Soda Lime):

$$R-COONa + NaOH \xrightarrow[\Delta]{\text{CaO (3:1)}} R-H + Na_2CO_3 \quad \text{(Product contains 1 fewer carbon atom)} \tag{3}$$

3. Kolbe Electrolytic Synthesis:

$$\text{At Anode: } 2R-COO^- \xrightarrow{-2e^-} 2R-COO^\bullet \xrightarrow{-2CO_2} 2R^\bullet \rightarrow R-R + 2CO_2$$

$$\text{At Cathode: } 2H_2O + 2e^- \rightarrow H_2\uparrow + 2OH^- \quad \text{(pH of solution increases during electrolysis)}$$

4. Free Radical Halogenation Reactivity & Selectivity:

$$\text{Reactivity: } F_2 > Cl_2 > Br_2 > I_2 \qquad \text{Selectivity: } 3^\circ\text{-H} > 2^\circ\text{-H} > 1^\circ\text{-H} \tag{4}$$

Relative rates of abstraction at 300°C:

- Chlorination ($1^\circ : 2^\circ : 3^\circ = 1 : 3.8 : 5.0$) → Low selectivity, governed by probability factor.
- Bromination ($1^\circ : 2^\circ : 3^\circ = 1 : 82 : 1600$) → Highly selective for 3° carbon.

Alkene Addition Reactions & Regiochemical Rules

1. Electrophilic Addition of HX (Markownikoff's Rule): The electrophile (H^+) adds to the double bond to form the more stable carbocation intermediate (allows rearrangement!).

$$CH_3 - CH = CH_2 + HCl \longrightarrow CH_3 - CH(Cl) - CH_3 \quad \text{(2-chloropropane)} \tag{5}$$

2. Peroxide Effect / Kharasch Effect (Anti-Markownikoff Addition): Occurs **only with** HBr in the presence of organic peroxides ($R - O - O - R$) via free radical mechanism.

$$CH_3 - CH = CH_2 + HBr \xrightarrow{\text{peroxide}} CH_3 - CH_2 - CH_2 - Br \quad \text{(1-bromopropane)} \tag{6}$$

Why only HBr*?* Addition of $Cl^\bullet$ to alkene is endothermic in step 2; abstraction of H from HI by radical is endothermic in step 1. Only with HBr are both propagation steps exothermic!

3. Hydroboration-Oxidation (Syn-Addition, Anti-Markownikoff Hydration):

$$6R - CH = CH_2 \xrightarrow{B_2H_6/THF} 2(R - CH_2 - CH_2)_3B \xrightarrow{H_2O_2/OH^-} 3R - CH_2 - CH_2 - OH + B(OH)_3 \tag{7}$$

Regioselectivity: Anti-Markownikoff alcohol; Stereochemistry: **Syn-addition**; No carbocation rearrangement!

4. Oxymercuration-Demercuration (Markownikoff Hydration without Rearrangement):

$$R - CH = CH_2 \xrightarrow[H_2O]{Hg(OAc)_2} R - CH(OH) - CH_2 - HgOAc \xrightarrow{NaBH_4/OH^-} R - CH(OH) - CH_3 \tag{8}$$

Proceeds via cyclic mercurinium ion $\implies$ strictly NO rearrangement.

5. Ozonolysis (Reductive vs Oxidative Cleavage):

$$R_2C = CHR' \xrightarrow{O_3/CH_2Cl_2} \text{Molar Ozonide} \xrightarrow[Zn/H_2O]{\text{Reductive}} R_2C = O + R'CHO + ZnO \tag{9}$$

If oxidative (H_2O_2 or Ag_2O), any aldehyde formed is oxidized to carboxylic acid ($R'COOH$).

Alkynes: Acidity, Hydration & Stereoselective Reduction

1. Terminal Alkyne Acidity: *sp*-hybridized carbon has 50% *s*-character, stabilizing conjugate carbanion:

$$R - C \equiv CH + NaNH_2 \longrightarrow R - C \equiv C^-Na^+ + NH_3 \uparrow$$
$$R - C \equiv CH + [Ag(NH_3)_2]^+ \longrightarrow R - C \equiv C - Ag \downarrow \quad \text{(White precipitate, Tollens test)}$$
$$R - C \equiv CH + Cu_2Cl_2 + NH_4OH \longrightarrow R - C \equiv C - Cu \downarrow \quad \text{(Red precipitate)}$$

2. Kucherov Reaction (Hydration of Alkynes):

$$HC \equiv CH + H_2O \xrightarrow[333\,K]{HgSO_4/\text{dil. } H_2SO_4} [CH_2 = CH - OH] \xrightarrow{\text{tautomerism}} CH_3 - CHO \quad \text{(Acetaldehyde)} \tag{10}$$

All other terminal and non-terminal alkynes yield ketones:

$$CH_3 - C \equiv CH + H_2O \xrightarrow{Hg^{2+}/H^+} CH_3 - CO - CH_3 \quad \text{(Acetone)} \tag{11}$$

3. Stereospecific Reduction of Alkynes:

- **Lindlar's Catalyst** ($H_2/Pd - CaCO_3/\text{quinoline}$)**:** Syn-addition $\rightarrow$ ***cis*-Alkene**.
- **Birch Reduction** ($Na/\text{liq. } NH_3$ **with EtOH):** Anti-addition $\rightarrow$ ***trans*-Alkene**.

Aromatic Hydrocarbons: EAS Directing Groups & Activating Power

Electrophilic Aromatic Substitution (EAS) involves arenium ion (σ-complex / Wheland intermediate):

Group Category	Examples	Directing	Reactivity vs Benzene
Strong Activators	$-OH, -O^-, -NH_2, -NHR, -NR_2$	*ortho / para*	Greatly increased
Moderate Activators	$-OR, -NHCOR, -OCOR$	*ortho / para*	Moderately increased
Weak Activators	$-CH_3, -C_2H_5, -Ar$	*ortho / para*	Slightly increased
Deactivating *o/p*-directors	$-F, -Cl, -Br, -I$	***ortho / para***	**Decreased ($-I > +M$)**
Moderate Deactivators	$-CHO, -COR, -COOH, -COOR, -SO_3H$	*meta*	Moderately decreased
Strong Deactivators	$-NO_2, -NR_3^+, -CF_3, -CCl_3, -CN$	*meta*	Greatly decreased

NEET Trap: Friedel-Crafts Reactions Limitations

1. **Aniline does NOT undergo Friedel-Crafts reaction:** The Lewis acid catalyst ($AlCl_3$) complexes with the lone pair on the $-NH_2$ group: $Ph-NH_2 + AlCl_3 \rightarrow Ph-\overset{+}{N}H_2-\bar{A}lCl_3$. The resulting positive charge strongly deactivates the aromatic ring.
2. **Strongly deactivated rings do not react:** Nitrobenzene, benzoic acid, and benzonitrile do not undergo alkylation or acylation (used as solvents).
3. **Polyalkylation vs Monoacylation:** Alkylation introduces an activating alkyl group, leading to polyalkylation. Acylation introduces a deactivating acyl group ($-COR$), so it strictly stops at monoacylation.
4. **Rearrangement in Alkylation:** 1-chloropropane + $Ph-H \xrightarrow{AlCl_3}$ isopropylbenzene (Cumene), due to $1^\circ \rightarrow 2^\circ$ carbocation hydride shift!

3 Haloalkanes & Haloarenes: S_N1, S_N2, $E1$, $E2$

Nucleophilic Substitution Mechanisms: S_N1 vs S_N2 Core Matrix

Parameter	S_N1 (Substitution Nucleophilic Unimolecular)	S_N2 (Substitution Nucleophilic Bimolecular)
Kinetics	First Order: Rate $= k[R-X]$	Second Order: Rate $= k[R-X][Nu^-]$
Mechanism	Two steps (carbocation intermediate)	One step concerted (transition state)
Reactivity of RX	$3^\circ > 2^\circ > 1^\circ > CH_3X$ (carbocation stability)	$CH_3X > 1^\circ > 2^\circ > 3^\circ$ (steric hindrance)
Stereochemistry	**Racemization** (partial net inversion)	**100% Walden Inversion**
Nucleophile	Weak nucleophile (H_2O, ROH); low concentration	Strong nucleophile (OH^-, RO^-, CN^-); high conc.
Solvent	**Polar Protic** (H_2O, EtOH, AcOH)	**Polar Aprotic** (DMSO, DMF, Acetone)
Rearrangement	**Possible** (hydride/alkyl shift)	**Strictly Impossible**
Substrate Limits	Allylic & Benzylic halides react rapidly	Vinyl ($CH_2=CHX$) and Aryl (PhX) unreactive

Ambident Nucleophiles in Haloalkane Reactions

Ambident nucleophiles possess two nucleophilic centers with differing electronegativities:

$$R-X + KCN \xrightarrow{\text{aq. EtOH}} R-CN \quad \text{(Ionic; C-attack forms stronger C-C bond)}$$

$$R-X + AgCN \xrightarrow{\text{aq. EtOH}} R-NC \quad \text{(Covalent; lone pair on N attacks)}$$

$$R-X + KNO_2 \xrightarrow{\text{aq. EtOH}} R-O-N=O \quad \text{(Ionic; negative charge on O attacks)}$$

$$R-X + AgNO_2 \xrightarrow{\text{aq. EtOH}} R-NO_2 \quad \text{(Covalent; lone pair on N attacks)}$$

Elimination Regiochemistry: Saytzeff vs Hofmann Product

- **Saytzeff (Zaitsev) Rule:** Base removes β-hydrogen from the β-carbon having fewer hydrogens, yielding the more substituted, thermodynamically more stable alkene (major product with small bases: OH^-, EtO^-, MeO^-).
- **Hofmann Rule:** Base removes β-hydrogen from the least hindered β-carbon, yielding the less substituted alkene as major product. Favored when:
  1. Attacking base is bulky (e.g., Potassium *tert*-butoxide: t-BuO^-K^+).
  2. Leaving group is poor (e.g., Fluoride $-F$, or quarternary ammonium $-NR_3^+$).

NEET Trap: Nucleophilic Inactivity of Haloarenes

Chlorobenzene (C_6H_5Cl) does not undergo nucleophilic substitution under normal conditions due to:

1. **Resonance:** Delocalization of chlorine lone pair gives partial double bond character to the $C-Cl$ bond (169 pm vs 177 pm in $R-Cl$).
2. **Hybridization of Carbon:** Chlorine is attached to an sp^2 hybridized carbon (33.3% s-character), which is more electronegative and holds electrons tighter than sp^3 carbon.
3. **Instability of Phenyl Cation:** Self-ionization to phenyl cation is energetically forbidden.
4. **Electrostatic Repulsion:** Electron-rich π-cloud of benzene repels incoming nucleophile.

Remedy (S_NAr Mechanism): Introduction of strong electron-withdrawing groups ($-NO_2$) at ***ortho* and *para* positions** stabilizes the Meisenheimer intermediate anionic complex, drastically decreasing required temperature:

- Chlorobenzene: NaOH, 623 K, 300 atm (Dow Process).
- p-Nitrochlorobenzene: NaOH, 443 K, H^+.
- 2,4-Dinitrochlorobenzene: NaOH, 368 K, H^+.
- 2,4,6-Trinitrochlorobenzene: Warm with H_2O alone at 323 K!

4 Oxygen Compounds: Alcohols, Phenols, Ethers, Carbonyls & Acids

Alcohols & Phenols Preparation & Conversions

1. Grignard Reagent (RMgX) Carbonyl Additions:

$$HCHO + RMgX \xrightarrow{H_3O^+} 1^\circ \text{ Alcohol} \qquad R'CHO + RMgX \xrightarrow{H_3O^+} 2^\circ \text{ Alcohol}$$

$$R'COR'' + RMgX \xrightarrow{H_3O^+} 3^\circ \text{ Alcohol} \qquad CO_2 + RMgX \xrightarrow{H_3O^+} RCOOH \quad \text{(Carboxylic acid)}$$

2. Cumene Hydroperoxide Process (Industrial Phenol):

$$\text{Benzene} + CH_3 - CH = CH_2 \xrightarrow{H_3PO_4} \text{Cumene} \xrightarrow{O_2} \text{Cumene Hydroperoxide} \xrightarrow{\text{dil. } H_2SO_4} \text{Phenol} + CH_3COCH_3 \quad \text{(Acetone by-product)}$$

3. Kolbe-Schmitt Reaction (Synthesis of Salicylic Acid):

$$\text{Sodium phenoxide} + CO_2 \xrightarrow[4-7\text{ atm}]{400\text{ K}} \text{Sodium salicylate} \xrightarrow{H^+} \text{Salicylic acid (2-hydroxybenzoic acid)} \tag{12}$$

4. Reimer-Tiemann Reaction (Synthesis of Salicylaldehyde):

$$\text{Phenol} + CHCl_3 + 3NaOH \xrightarrow{340\text{ K}} \text{Salicylaldehyde} \qquad (\textbf{Electrophile: Dichlorocarbene} : CCl_2) \tag{13}$$

Note: If CCl_4 is used instead of $CHCl_3$, salicylic acid is the main product.

5. Esterification (Fischer):

$$RCOOH + R'OH \underset{}{\overset{\text{conc. } H_2SO_4}{\rightleftharpoons}} RCOOR' + H_2O \quad \text{(Reactivity of ROH: } CH_3OH > 1^\circ > 2^\circ > 3^\circ) \tag{14}$$

Ethers: Williamson Synthesis & Cleavage with HI

1. Williamson Ether Synthesis:

$$R - X + R' - O^-Na^+ \longrightarrow R - O - R' + NaX \quad (\textbf{Mechanism: Strictly } S_N2) \tag{15}$$

- Alkyl halide must be 1° (or CH_3X).
- If 3° alkyl halide is used with alkoxide, **elimination ($E2$) dominates exclusively**:

$$(CH_3)_3C - Br + CH_3O^-Na^+ \longrightarrow CH_3 - C(CH_3) = CH_2 + CH_3OH + NaBr \quad \text{(Isobutylene)} \tag{16}$$

2. Cleavage of Ethers by Halogen Acid (HI):

- **Alkyl Alkyl Ethers ($1^\circ/2^\circ$ groups):** Follows S_N2. Iodine ion attacks the less sterically hindered alkyl group:

$$CH_3 - O - C_2H_5 + HI \xrightarrow{\text{cold}} CH_3 - I + C_2H_5 - OH \tag{17}$$

- **One group is 3° (or Benzyl):** Follows S_N1 via stable carbocation; iodine attacks the 3° carbon:

$$(CH_3)_3C - O - CH_3 + HI \xrightarrow{\text{cold}} (CH_3)_3C - I + CH_3 - OH \tag{18}$$

- **Alkyl Aryl Ethers (Anisole):** The O − Aryl bond has partial double bond character and never cleaves:

$$C_6H_5 - O - CH_3 + HI \xrightarrow{\Delta} C_6H_5 - OH \text{ (Phenol)} + CH_3 - I \tag{19}$$

Carbonyl Compounds: Nucleophilic Addition & Carbonyl Condensations

1. Relative Reactivity Towards Nucleophilic Addition:

$$HCHO > CH_3CHO > CH_3COCH_3 > PhCHO > PhCOCH_3 > PhCOPh \tag{20}$$

Governed by: Steric hindrance at carbonyl carbon and $+I/+M$ electron donation reducing electrophilicity of carbonyl carbon.

2. Aldol Condensation (α-H Required): Carbonyl compounds with at least one α-hydrogen react in dilute alkali (10% NaOH or $Ba(OH)_2$):

$$2CH_3CHO \xrightarrow{\text{dil. NaOH}} CH_3 - CH(OH) - CH_2 - CHO \xrightarrow[\Delta]{-H_2O} CH_3 - CH = CH - CHO \quad \text{(Crotonaldehyde)} \tag{21}$$

3. Cannizzaro Reaction (No α-H Present): Self-oxidation and self-reduction in concentrated alkali (50% NaOH):

$$2HCHO + \text{conc. NaOH} \longrightarrow CH_3OH \text{ (Methanol)} + HCOONa \text{ (Sodium formate)} \tag{22}$$

$$2C_6H_5CHO + \text{conc. NaOH} \longrightarrow C_6H_5CH_2OH \text{ (Benzyl alcohol)} + C_6H_5COONa \text{ (Sodium benzoate)} \tag{23}$$

Cross Cannizzaro with Formaldehyde: Formaldehyde undergoes oxidation exclusively ($HCOO^-$) because hydride transfer occurs from the more nucleophilically attacked formaldehyde.

Named Carbonyl Reductions, Oxidations & Acid Reactions

1. Clemmensen Reduction: $Zn - Hg + \text{conc. HCl}$ $(> C = O \rightarrow > CH_2)$; *Avoid if acid-sensitive groups (e.g. $-OH, -C=C-, -OR$) present.*

2. Wolff-Kishner Reduction: $NH_2 - NH_2 + KOH/\text{glycol}, \ \Delta$ $(> C = O \rightarrow > CH_2)$; *Avoid if base-sensitive groups present.*

3. Rosenmund Reduction: $R - COCl + H_2 \xrightarrow{Pd/BaSO_4, \text{quinoline}} R - CHO + HCl.$

4. Stephen Reaction: $R - CN + SnCl_2/HCl \longrightarrow R - CH = NH \cdot HCl \xrightarrow{H_3O^+} R - CHO + NH_4Cl.$

5. Etard Reaction: $C_6H_5 - CH_3 + CrO_2Cl_2/CS_2 \rightarrow$ Chromium complex $\xrightarrow{H_3O^+} C_6H_5 - CHO.$

6. Gattermann-Koch Reaction: $C_6H_6 + CO + HCl \xrightarrow{\text{anhyd. } AlCl_3/CuCl} C_6H_5 - CHO + HCl.$

7. Hell-Volhard-Zelinsky (HVZ) Reaction: Carboxylic acids with α-hydrogen react with halogen in presence of red phosphorus:

$$R - CH_2 - COOH + X_2 \xrightarrow[\text{(ii) } H_2O]{\text{(i) Red P}} R - CH(X) - COOH \quad (\alpha\text{-halocarboxylic acid; X = Cl, Br}) \tag{24}$$

5 Nitrogen Compounds: Amines & Diazonium Salts

Synthesis & Reactions of Amines

1. Gabriel Phthalimide Synthesis (Pure 1° Aliphatic Amines):

$$\text{Phthalimide} \xrightarrow{\text{KOH}} \text{K-phthalimide} \xrightarrow{\text{R}-\text{X}\,(S_N2)} N\text{-Alkylphthalimide} \xrightarrow{\text{NaOH(aq)}} \text{R}-\text{NH}_2 + \text{Phthalate}$$

Limitation: Aromatic primary amines cannot be prepared because aryl halides do not undergo S_N2 displacement.

2. Hoffmann Bromamide Degradation: Conversion of an amide to a 1° amine containing **one carbon less**:

$$\text{R}-\text{CONH}_2 + \text{Br}_2 + 4\text{NaOH} \longrightarrow \text{R}-\text{NH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O} \tag{25}$$

Intermediate is alkyl isocyanate ($\text{R}-\text{N}=\text{C}=\text{O}$); migration of alkyl group occurs with **retention of configuration**.

3. Carbylamine Test (Isocyanide Test - Strictly 1° Amines): Both aliphatic and aromatic 1° amines react with chloroform and alc. KOH to form foul-smelling isocyanides:

$$\text{R}-\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \xrightarrow{\Delta} \text{R}-\text{NC}\ (\text{Alkyl isocyanide}) + 3\text{KCl} + 3\text{H}_2\text{O} \tag{26}$$

2° and 3° amines do NOT show this test.

4. Hinsberg Test (Distinction of 1°, 2°, 3° Amines): Reagent: Benzenesulfonyl chloride ($\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$).

- 1° Amine → *N*-alkylbenzenesulfonamide → possesses acidic H on N → **Soluble in alkali** (KOH).
- 2° Amine → *N*,*N*-dialkylbenzenesulfonamide → no acidic H → **Insoluble in alkali**.
- 3° Amine → No replaceable H on N → **Does not react with Hinsberg reagent**.

Diazonium Chemistry: Arenediazonium Salt Reactions

Diazotization: $\text{Ar}-\text{NH}_2 + \text{NaNO}_2 + 2\text{HCl} \xrightarrow{273-278\,\text{K}} \text{Ar}-\text{N}_2^+\text{Cl}^- + \text{NaCl} + 2\text{H}_2\text{O}.$

$$\begin{aligned}
\text{Sandmeyer:}\ & \text{Ar}-\text{N}_2^+\text{Cl}^- \xrightarrow{\text{Cu}_2\text{X}_2/\text{HX}} \text{Ar}-\text{X} \quad (\text{X} = \text{Cl}, \text{Br}, \text{CN}) \\
\text{Gattermann:}\ & \text{Ar}-\text{N}_2^+\text{Cl}^- \xrightarrow{\text{Cu/HX}} \text{Ar}-\text{X} \quad (\text{X} = \text{Cl}, \text{Br}) \\
\text{Balz-Schiemann:}\ & \text{Ar}-\text{N}_2^+\text{Cl}^- \xrightarrow{\text{HBF}_4} \text{Ar}-\text{N}_2^+\text{BF}_4^- \xrightarrow{\Delta} \text{Ar}-\text{F} + \text{BF}_3 + \text{N}_2\uparrow \\
\text{Iodination:}\ & \text{Ar}-\text{N}_2^+\text{Cl}^- + \text{KI} \xrightarrow{\text{warm}} \text{Ar}-\text{I} + \text{KCl} + \text{N}_2\uparrow \\
\text{Reduction:}\ & \text{Ar}-\text{N}_2^+\text{Cl}^- + \text{H}_3\text{PO}_2 + \text{H}_2\text{O} \xrightarrow{\text{Cu}^+} \text{Ar}-\text{H} + \text{H}_3\text{PO}_3 + \text{HCl} + \text{N}_2\uparrow \\
& \text{Ar}-\text{N}_2^+\text{Cl}^- + \text{CH}_3\text{CH}_2\text{OH} \longrightarrow \text{Ar}-\text{H} + \text{CH}_3\text{CHO} + \text{HCl} + \text{N}_2\uparrow \\
\text{Hydrolysis:}\ & \text{Ar}-\text{N}_2^+\text{Cl}^- + \text{H}_2\text{O} \xrightarrow{373\,\text{K}} \text{Ar}-\text{OH} + \text{HCl} + \text{N}_2\uparrow
\end{aligned}$$

Azo Coupling Reactions (Formation of Dyes):

$$\begin{aligned}
\text{With Phenol:}\ & \text{Ph}-\text{N}_2^+\text{Cl}^- + \text{C}_6\text{H}_5-\text{OH} \xrightarrow{\text{pH}\approx 9-10} p\text{-hydroxyazobenzene (Orange dye)} \\
\text{With Aniline:}\ & \text{Ph}-\text{N}_2^+\text{Cl}^- + \text{C}_6\text{H}_5-\text{NH}_2 \xrightarrow{\text{pH}\approx 4-5} p\text{-aminoazobenzene (Yellow dye)}
\end{aligned}$$

6 Biomolecules: Carbohydrates, Amino Acids & Nucleic Acids

Carbohydrates Classification & Reactions of D-Glucose

1. Classification & Reducing Nature:

- **Reducing Sugars:** All monosaccharides (aldoses & ketoses) and disaccharides with a free hemiacetal/hemiketal -OH at anomeric carbon (Maltose, Lactose). Reduce Tollens and Fehling solutions.
- **Non-Reducing Sugars:** Glycosidic linkage involves anomeric carbons of both units (e.g., **Sucrose** = α-D-glucopyranosyl-(1→2)-β-D-fructofuranoside).

- **Invert Sugar:** Sucrose ($[\alpha]_D = +66.5°$) on hydrolysis gives D-glucose ($+52.5°$) and D-fructose ($-92.4°$). The mixture is levorotatory ($-39.9°/2 = -19.9°$).

2. Chemical Reactions of D-Glucose ($C_6H_{12}O_6$):

$$\begin{aligned}
\text{Glucose} + \text{HI},\ \Delta &\longrightarrow n\text{-hexane} \quad \text{(Proves open chain of 6 carbons)} \\
\text{Glucose} + NH_2OH &\longrightarrow \text{Glucose oxime} \quad \text{(Confirms carbonyl group } > C = O) \\
\text{Glucose} + HCN &\longrightarrow \text{Glucose cyanohydrin} \\
\text{Glucose} + Br_2/H_2O &\longrightarrow \text{Gluconic acid (mild oxidation of } -CHO \text{ to } -COOH) \\
\text{Glucose} + \text{conc. } HNO_3 &\longrightarrow \text{Saccharic acid / Glucaric acid (oxidizes both } -CHO \text{ and } 1°\text{-OH}) \\
\text{Glucose} + 5(CH_3CO)_2O &\longrightarrow \text{Glucose pentaacetate} \quad \text{(Confirms presence of 5 independent } -OH \text{ groups)}
\end{aligned}$$

3. Why the Cyclic Structure of Glucose? D-Glucose does not give Schiff's test, 2,4-DNP test, and does not form hydrogen sulphite addition product with $NaHSO_3$. The pentaacetate does not react with hydroxylamine (absence of free $-CHO$).

$$\alpha\text{-D-(+)-glucose } ([\alpha]_D = +112°) \underset{\text{in solution}}{\overset{\text{Mutarotation}}{\rightleftharpoons}} \text{Equilibrium } (+52.7°) \overset{\beta}{\rightleftharpoons} \text{-D-(+)-glucose } (+19°) \tag{27}$$

α and β isomers differ only at $C-1$ (anomeric carbon) $\implies$ **Anomers**.

Amino Acids, Peptide Bonds & Protein Structure

1. Zwitterion & Isoelectric Point (pI): In aqueous solution, the carboxyl group loses a proton and amino group accepts it:

$$R - CH(NH_2) - COOH \rightleftharpoons R - CH(NH_3^+) - COO^- \quad \text{(Dipolar Zwitterion; net charge } = 0) \tag{28}$$

$$\text{At pH} < \text{pI}: \text{ exists as cation } R - CH(NH_3^+) - COOH \quad \text{(migrates towards cathode).} \tag{29}$$

$$\text{At pH} > \text{pI}: \text{ exists as anion } R - CH(NH_2) - COO^- \quad \text{(migrates towards anode).} \tag{30}$$

For neutral amino acid: $\text{pI} = \frac{pK_{a1} + pK_{a2}}{2}$.

- **Chirality:** All naturally occurring amino acids are optically active and possess **L-configuration**, **except Glycine** ($H_2N - CH_2 - COOH$, achiral).
- **Essential Amino Acids (10):** Valine, Leucine, Isoleucine, Phenylalanine, Methionine, Tryptophan, Threonine, Lysine, Histidine, Arginine. [**Mnemonic: PVT TIM HALL**].

2. Protein Structural Levels:

- **Primary:** Specific sequence of amino acids linked by covalent peptide bonds ($-CO - NH-$).
- **Secondary:** Local conformation of polypeptide chain stabilized by **hydrogen bonds** between $> C = O$ and $-NH-$ groups: α-helix (intramolecular H-bonds) and β-pleated sheets (intermolecular H-bonds).
- **Tertiary:** Overall 3D folding of polypeptide chains stabilized by H-bonds, disulphide linkages ($-S - S-$), van der Waals forces, and electrostatic attractions.
- **Quaternary:** Spatial arrangement of two or more subunits (e.g., Hemoglobin with 2α and 2β chains).
- **Denaturation:** Heat, pH change breaks $2°, 3°, 4°$ structures; **Primary structure remains completely intact**! (e.g., coagulation of egg white, curdling of milk).

Nucleic Acids: Nucleotides, Chargaff's Rules & Double Helix

1. Structural Components:

- **Pentose Sugar:** β-D-ribose (RNA) vs β-D-2-deoxyribose (DNA).
- **Nitrogenous Bases:**
  - **Purines (Bicyclic):** Adenine (A), Guanine (G).
  - **Pyrimidines (Monocyclic):** Cytosine (C), Thymine (T, DNA only), Uracil (U, RNA only).
- **Nucleoside:** Sugar + Base (attached at $C-1'$ of sugar via β-glycosidic linkage).
- **Nucleotide:** Nucleoside + Phosphate (esterified at $C-5'$ of sugar).
- **Phosphodiester Linkage:** Connects $C-3'$ of one nucleotide to $C-5'$ of adjacent nucleotide.

2. Chargaff's Rules (for Double-Stranded DNA):

$$\%A = \%T \quad \text{and} \quad \%G = \%C \implies \frac{A+G}{T+C} = 1 \qquad \text{(Purines = Pyrimidines)} \tag{31}$$

Base pairing: A pairs with T via **2 hydrogen bonds** (A = T); G pairs with C via **3 hydrogen bonds** (G ≡ C).

7 Master Identification Tests & Distinction Matrix

Comprehensive Tabular Matrix for Distinguishing Organic Functional Pairs

Compound Pair	Reagent / Test	Observed Visual Distinction
$1^\circ, 2^\circ, 3^\circ$ Alcohols	Lucas Reagent (conc. HCl + anhyd. $ZnCl_2$)	3°: immediate cloudiness/turbidity; 2°: turbidity in 5 minutes; 1°: no turbidity at room temperature.
Aliphatic vs Aromatic Aldehydes	Fehling's Solution ($A + B$)	Aliphatic: Red ppt of Cu_2O; Benzaldehyde: No reaction (does not reduce Fehling).
Aldehyde vs Ketone	Tollens' Reagent ($[Ag(NH_3)_2]^+$)	Aldehydes: Bright silver mirror (Ag); Ketones: Negative.
$CH_3 - C = O$ or $CH_3 - CH(OH) -$ compounds	Iodoform Test ($I_2 + NaOH$)	Yellow crystalline precipitate of CHI_3 (m.p. 119°C) with antiseptic odor. Positive for ethanol, acetaldehyde, 2-propanol, methyl ketones.
Carboxylic Acid vs Phenol	$NaHCO_3$ (aqueous)	Carboxylic acid: Brisk effervescence of CO_2; Phenol: No effervescence.
Phenol vs Alcohol	Neutral $FeCl_3$ solution	Phenol: Characteristic violet/purple coloration ($[Fe(OC_6H_5)_6]^{3-}$); Alcohols: Negative.
1° Amine vs $2^\circ/3^\circ$	Carbylamine Test ($CHCl_3$ + alc. KOH)	Only 1° amine gives extremely foul smell of isocyanide ($R - NC$).
$1^\circ, 2^\circ, 3^\circ$ Amines	Hinsberg Reagent ($C_6H_5SO_2Cl$)	1°: Soluble in KOH; 2°: Precipitate insoluble in KOH; 3°: No reaction.
Terminal vs Internal Alkyne	Ammoniacal Cu_2Cl_2	Terminal ($R - C \equiv CH$): Red ppt of copper acetylide; Internal: Negative.
Alkene/Alkyne vs Alkane	Bromine water (Br_2/CCl_4)	Unsaturated: Decolorization of reddish-brown bromine color without evolution of HBr.

High-Yield Reagent Conversion Master Reference

- PCC **(Pyridinium Chlorochromate) / Collins Reagent:** Oxidizes 1° alcohols strictly to aldehydes (RCHO) without over-oxidation to carboxylic acid. Oxidizes 2° alcohols to ketones.
- **DIBAL-H (Diisobutylaluminium Hydride):** Selectively reduces esters ($RCOOR'$) and nitriles (RCN) at low temperature (-78°C) to **aldehydes** (RCHO).
- $LiAlH_4$ **vs** $NaBH_4$**:** $LiAlH_4$ reduces acids, esters, amides, nitriles, aldehydes, and ketones. $NaBH_4$ is weaker and selectively reduces **only aldehydes, ketones, and acid halides** (does not touch esters, acids, or amides).
- CrO_3 **in glacial acetic acid vs Jones Reagent** (CrO_3/H_2SO_4)**:** Mild vs strong chromic acid oxidation.
- **Fenton's Reagent** ($FeSO_4 + H_2O_2$)**:** Generates hydroxyl free radicals ($OH^\bullet$) for selective oxidation of alcohols to carbonyls.